

Reproduce-then-Extend — Building Energy Efficiency (Engineering)

(Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Tsanas, A. & Xifara, A. (2012). “Accurate quantitative estimation of energy performance of residential buildings using statistical machine learning tools.” *Energy and Buildings* 49:560–567. The paper’s baseline is a **linear regression** predicting a building’s heating load from 8 geometry inputs ($n = 768$; UCI id 242). This example illustrates the first regression extension: **polynomial terms**.

1 Background

Tsanas and Xifara study how a building’s **energy performance** depends on its **design geometry**, to help architects design more efficient buildings. Using 768 simulated building shapes, they estimate the **heating (and cooling) load** from eight geometry inputs, using a **linear regression** as the interpretable baseline alongside nonlinear machine-learning tools. The **primary conclusion** is that energy load is accurately predictable from geometry – overall height, relative compactness, and glazing are the strongest drivers – and that flexible learners (random forests) substantially outperform the linear model, motivating their use for this strongly nonlinear relationship.

2 Data and codebook

Unit of analysis: a simulated residential **building design** (12 shapes and variants, via Ecotect); $n = 768$.

Variable	Definition
Y1	heating load (kWh per m^2 of floor area) (outcome)
X1	relative compactness (volume-to-surface ratio, normalized)
X2	surface area (m^2)
X3	wall area (m^2)
X4	roof area (m^2)
X5	overall height (m)
X6	orientation (2/3/4/5 = N/E/S/W)
X7	glazing area (fraction of floor area)
X8	glazing-area distribution (0-5 scheme)

3 Descriptive analysis

Before fitting anything, we profile the data thoroughly. The goal is to understand the outcome’s shape, each geometry input’s distribution, how the inputs relate to heating load, and — critically for a regression — how the inputs relate to *each other*.

```

tmp_xlsx <- tempfile(fileext = ".xlsx")
download.file("https://raw.githubusercontent.com/bdesmarais/intro-ml-statistical-horizons-4day/main/tutorial/tutorial_data.xlsx", tmp_xlsx, mode = "wb", quiet = TRUE)
d <- as.data.frame(read_excel(tmp_xlsx)) # data ships in the tutorial folder
d <- d[, colSums(!is.na(d)) > 0]; d <- d[complete.cases(d), ]
preds <- paste0("X", 1:8)

```

3.1 Dataset and provenance

The **unit of analysis** is a *simulated residential building design*. Angeliki Xifara generated 12 base building shapes in the Ecotect simulator and varied their geometry (glazing, orientation, etc.), producing **n = 768** distinct designs. For each design a physics simulation reports the annual **heating load** (Y1) and cooling load (Y2); this tutorial models heating load. The data are deterministic simulation output, not a survey sample, so there is **no sampling error and no missingness** — every input combination is fully observed. The predictors are 8 geometry variables. The data come from the UCI Machine Learning Repository (id 242), released with Tsanas & Xifara (2012).

```

dict <- data.frame(
  Variable = c("Y1", paste0("X", 1:8)),
  Role = c("outcome", rep("predictor", 8)),
  Type = c("numeric", "numeric", "numeric", "numeric", "numeric",
           "binary (2 levels)", "categorical (4 orient.)",
           "ordinal (4 levels)", "ordinal (6 levels)"),
  Description = c(
    "heating load (kWh per m^2 floor)",
    "relative compactness (volume/surface, normalized)",
    "surface area (m^2)", "wall area (m^2)", "roof area (m^2)",
    "overall height (3.5 or 7 m)", "orientation (2/3/4/5 = N/E/S/W)",
    "glazing area (0, .1, .25, .4 of floor)",
    "glazing distribution (0-5 scheme)")
)
kable(dict, caption = "Variable dictionary: outcome and predictors.")

```

Table 1: Variable dictionary: outcome and predictors.

Variable	Role	Type	Description
Y1	outcome	numeric	heating load (kWh per m ² floor)
X1	predictor	numeric	relative compactness (volume/surface, normalized)
X2	predictor	numeric	surface area (m ²)
X3	predictor	numeric	wall area (m ²)
X4	predictor	numeric	roof area (m ²)
X5	predictor	binary (2 levels)	overall height (3.5 or 7 m)
X6	predictor	categorical (4 orient.)	orientation (2/3/4/5 = N/E/S/W)
X7	predictor	ordinal (4 levels)	glazing area (0, .1, .25, .4 of floor)
X8	predictor	ordinal (6 levels)	glazing distribution (0-5 scheme)

Although X1-X5 are stored as continuous numbers, the simulation grid means several take only a handful of distinct values (X5 is effectively binary: 3.5 m or 7 m; X6-X8 are categorical design choices). We treat them numerically to match the published baseline, but keep their discreteness in mind when reading the plots.

3.2 The outcome in depth

```

ystat <- function(x) c(
  n = length(x), mean = mean(x), sd = sd(x), min = min(x),
  Q1 = quantile(x, .25), median = median(x), Q3 = quantile(x, .75),
  max = max(x), IQR = IQR(x),
  skew = e1071::skewness(x), kurtosis = e1071::kurtosis(x))
kable(t(round(ystat(d$Y1), 2)),
  caption = "Distribution of heating load (Y1).")

```

Table 2: Distribution of heating load (Y1).

n	mean	sd	min	Q1.25%	median	Q3.75%	max	IQR	skew	kurtosis
768	22.31	10.09	6.01	12.99	18.95	31.67	43.1	18.68	0.36	-1.25

```

p_hist <- ggplot(d, aes(Y1)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
    fill = "#a6cee3", colour = "white") +
  geom_density(colour = "#1f78b4", linewidth = 0.7) +
  labs(x = "heating load (Y1)", y = "density", title = "Distribution")
p_box <- ggplot(d, aes(y = Y1)) +
  geom_boxplot(fill = "#a6cee3", width = 0.4) +
  labs(y = "heating load (Y1)", x = NULL, title = "Boxplot") +
  theme(axis.text.x = element_blank())
p_hist + p_box + plot_layout(widths = c(2.4, 1))

```

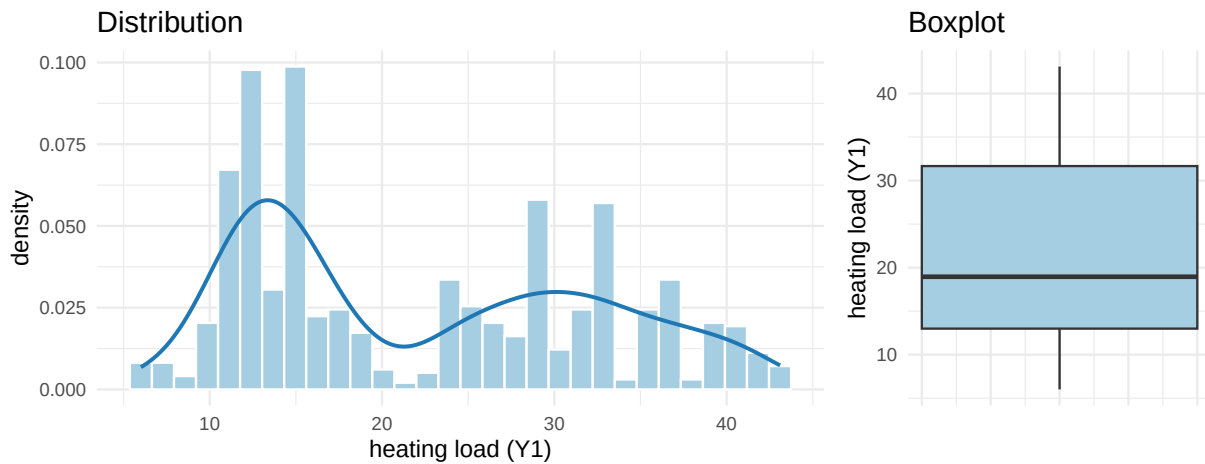


Figure 1: Heating load: histogram with density overlay (left) and boxplot (right). The distribution is broad and bimodal rather than a single bell.

Heating load ranges from about 6 to 43 kWh/m² (mean 22.3, sd 10.1). Skewness is only 0.36 — nearly symmetric — but the negative excess kurtosis (-1.25) and the histogram reveal a **bimodal, plateau-shaped** distribution, not a bell. This is a fingerprint of the design: buildings cluster into low-load and high-load regimes driven mostly by overall height. There is no zero-inflation, floor/ceiling piling, or long tail, so **no outcome transformation is warranted** — the spread is structural, and the regression's job is to explain it with the geometry inputs.

3.3 Univariate predictor distributions

```
psum <- function(x) c(
  n = length(x), n_miss = sum(is.na(x)), mean = mean(x), sd = sd(x),
  min = min(x), Q1 = quantile(x, .25), median = median(x),
  Q3 = quantile(x, .75), max = max(x), skew = e1071::skewness(x),
  n_unique = length(unique(x)))
ptab <- t(sapply(d[preds], psum))
kable(round(ptab, 2),
  caption = "Univariate summary of the 8 geometry predictors.")
```

Table 3: Univariate summary of the 8 geometry predictors.

	n	n_miss	mean	sd	min	Q1.25%	median	Q3.75%	max	skew	n_unique
X1	768	0	0.76	0.11	0.62	0.68	0.75	0.83	0.98	0.49	12
X2	768	0	671.71	88.09	514.50	606.38	673.75	741.12	808.50	-0.12	12
X3	768	0	318.50	43.63	245.00	294.00	318.50	343.00	416.50	0.53	7
X4	768	0	176.60	45.17	110.25	140.88	183.75	220.50	220.50	-0.16	4
X5	768	0	5.25	1.75	3.50	3.50	5.25	7.00	7.00	0.00	2
X6	768	0	3.50	1.12	2.00	2.75	3.50	4.25	5.00	0.00	4
X7	768	0	0.23	0.13	0.00	0.10	0.25	0.40	0.40	-0.06	4
X8	768	0	2.81	1.55	0.00	1.75	3.00	4.00	5.00	-0.09	6

```
d %>%
  select(all_of(preds)) %>%
  pivot_longer(everything(), names_to = "var", values_to = "value") %>%
  ggplot(aes(value)) +
  geom_histogram(bins = 20, fill = "#b2df8a", colour = "white") +
  facet_wrap(~ var, scales = "free", ncol = 4) +
  labs(x = NULL, y = "count")
```

The summary table exposes the grid structure: X5 (height) takes exactly two values, X6-X8 a handful, and even the “continuous” X1-X4 take 4-12 distinct levels. All predictors are close to symmetric ($|\text{skew}| < 0.6$), so no predictor transformations are indicated. The near-uniform spikes reflect a **balanced factorial design** — each geometry level appears in roughly equal numbers of buildings, which is ideal for isolating individual effects (were it not for the collinearity documented below).

3.4 Missing-data analysis

```
miss <- data.frame(
  variable = c("Y1", preds),
  n_missing = sapply(d[c("Y1", preds)], function(x) sum(is.na(x))),
  pct_missing = round(100 * sapply(d[c("Y1", preds)],
    function(x) mean(is.na(x))), 1))
kable(miss, row.names = FALSE,
  caption = "Missing values per variable.")
```

Table 4: Missing values per variable.

variable	n_missing	pct_missing
Y1	0	0

variable	n_missing	pct_missing
X1	0	0
X2	0	0
X3	0	0
X4	0	0
X5	0	0
X6	0	0
X7	0	0
X8	0	0

There is **no missing data**: all 768 rows are complete across all 9 variables. Because these are exhaustive simulation outputs over a designed grid, missingness is impossible by construction — the `complete.cases()` filter in the loading chunk is a safeguard that removes nothing here. The reproduction therefore uses the full $n = 768$ with no imputation or listwise deletion.

3.5 Bivariate relationships with the outcome

```
marg <- data.frame(
  predictor = preds,
  r_with_Y1 = round(sapply(d[preds], function(x) cor(x, d$Y1)), 3))
marg <- marg[order(-abs(marg$r_with_Y1)), ]
kable(marg, row.names = FALSE,
  caption = "Marginal (Pearson) correlation of each input with Y1.")
```

Table 5: Marginal (Pearson) correlation of each input with Y1.

predictor	r_with_Y1
X5	0.889
X4	-0.862
X2	-0.658
X1	0.622
X3	0.456
X7	0.270
X8	0.087
X6	-0.003

```
d %>%
  select(all_of(preds), Y1) %>%
  pivot_longer(all_of(preds), names_to = "var", values_to = "value") %>%
  ggplot(aes(value, Y1)) +
  geom_point(alpha = 0.15, colour = "#333333", size = 0.6) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE,
    colour = "#e31a1c", linewidth = 0.6) +
  facet_wrap(~ var, scales = "free_x", ncol = 4) +
  labs(x = NULL, y = "heating load (Y1)")
```

Four inputs carry almost all of the marginal signal. **Overall height** (X5, $r = 0.89$) and **relative compactness** (X1, $r = 0.62$) are the strongest positive drivers, while **roof area** (X4, $r = -0.86$) and **surface area** (X2, $r = -0.66$) pull load down. Orientation (X6) and glazing distribution (X8) are essentially uncorrelated with load — consistent with the physics that *how much* glazing and *how tall* a building is matter far more than which way

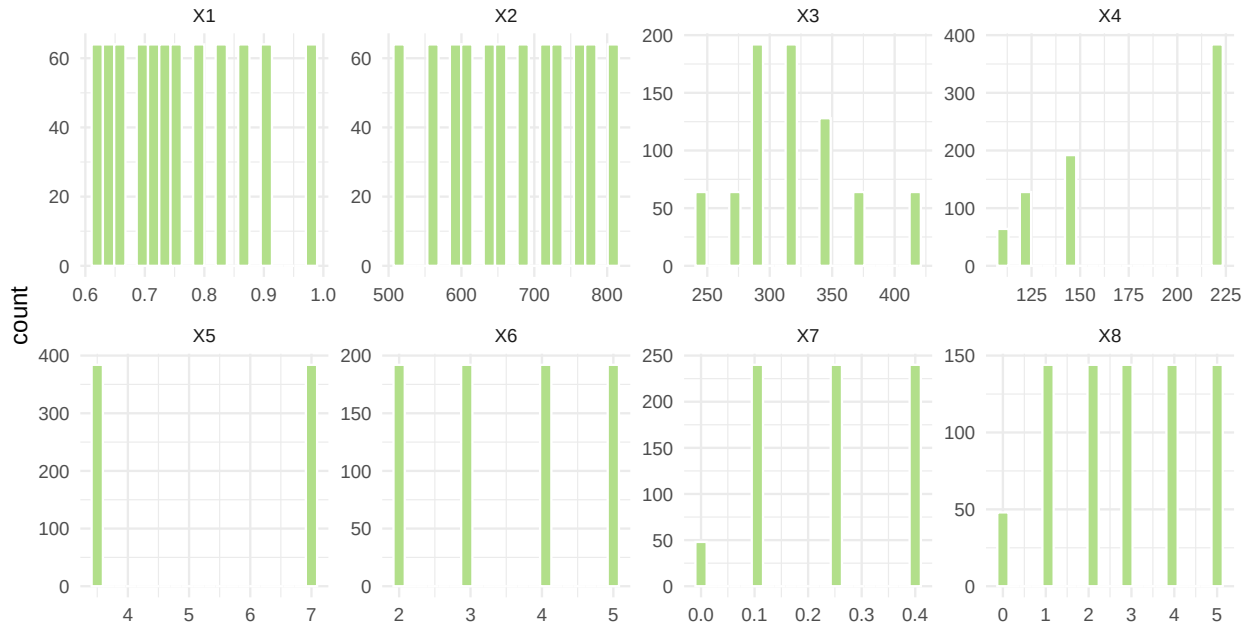


Figure 2: Distribution of each geometry input. Several inputs take only a few discrete values (the simulation grid), so they appear as spikes rather than smooth curves.

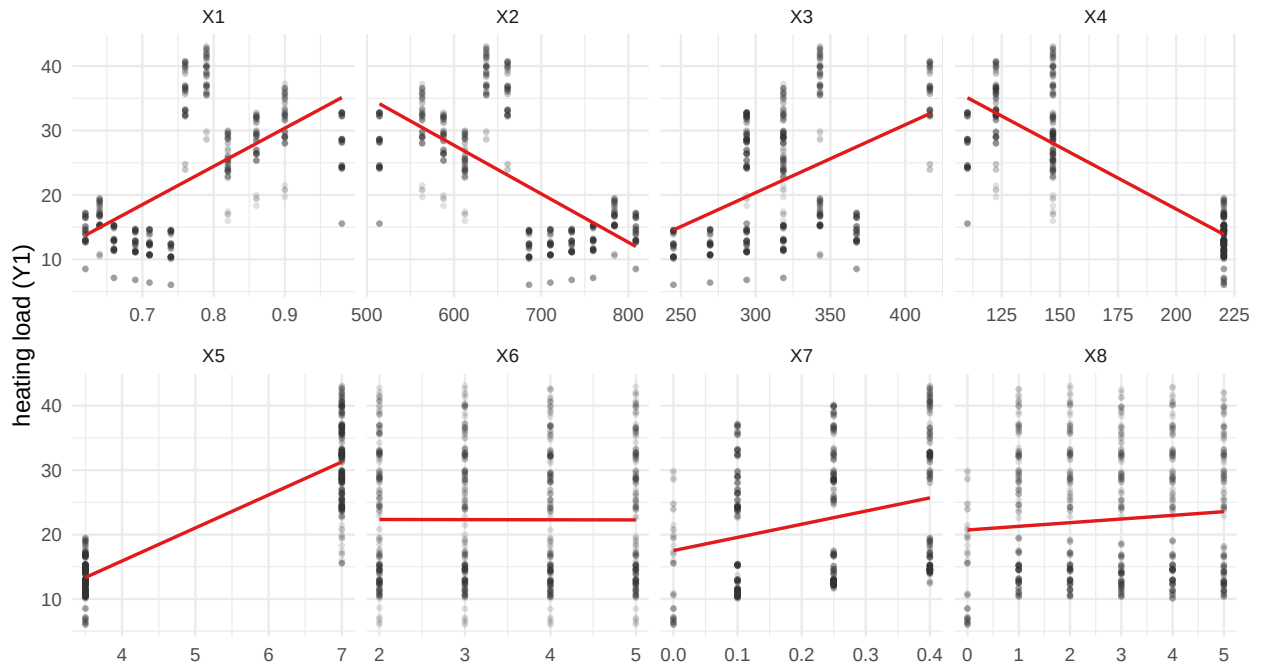


Figure 3: Heating load vs each geometry input, with a linear smoother. Height (X5) and compactness (X1) drive load up; surface and roof area (X2, X4) drive it down.

it faces. Several scatterplots (notably X1) show visible curvature around the linear smoother, foreshadowing the polynomial extension below.

3.6 Multivariate structure and collinearity

```
cmat <- cor(d[preds])
cmat[lower.tri(cmat)] <- NA
as.data.frame(cmat) %>%
  mutate(v1 = rownames(cmat)) %>%
  pivot_longer(-v1, names_to = "v2", values_to = "r") %>%
  filter(!is.na(r)) %>%
  ggplot(aes(v2, v1, fill = r)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", r)), size = 2.4) +
  scale_fill_gradient2(low = "#e31a1c", mid = "white",
                      high = "#1f78b4", midpoint = 0, limits = c(-1, 1)) +
  labs(x = NULL, y = NULL) +
  theme(panel.grid = element_blank())
```

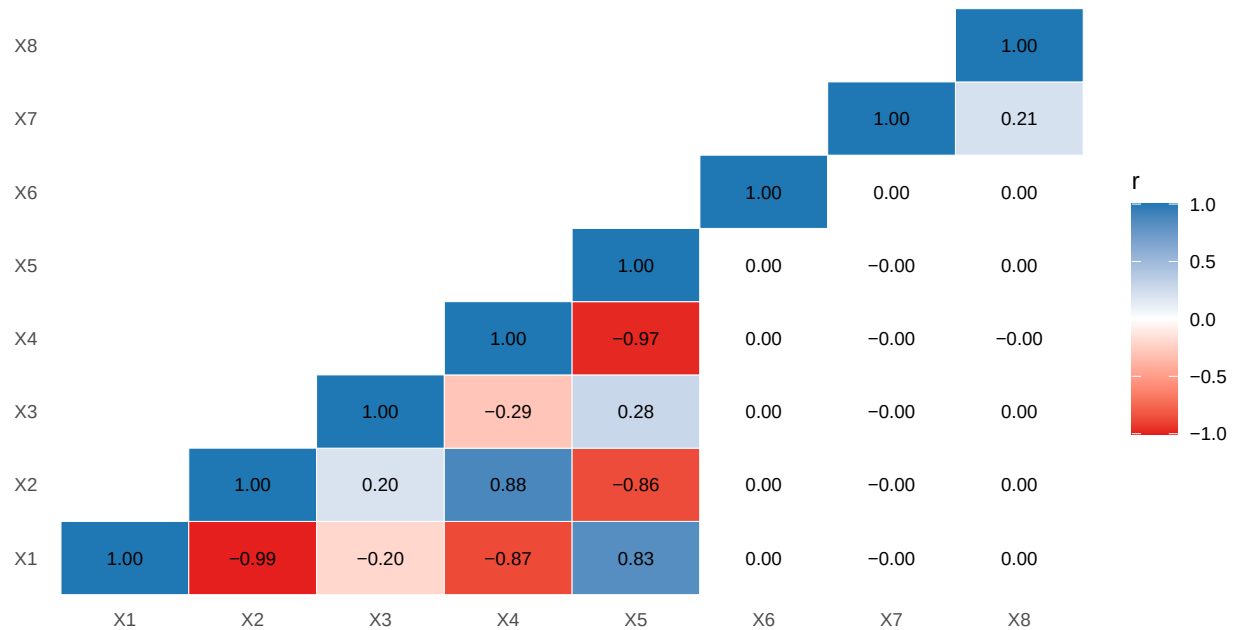


Figure 4: Correlation heatmap of the 8 geometry inputs. Deep red / blue cells mark $|r| > 0.7$ — the geometry variables are heavily interdependent.

```
X <- model.matrix(~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8, d)[, -1]
vif <- sapply(colnames(X), function(v) {
  r2 <- summary(lm(X[, v] ~ X[, setdiff(colnames(X), v)]))$r.squared
  1 / (1 - r2)
})
kable(t(round(vif, 1)),
      caption = "Variance inflation factors from the reproduction model.")
```

Table 6: Variance inflation factors from the reproduction model.

X1	X2	X3	X4	X5	X6	X7	X8
105.5	Inf	Inf	Inf	31.2	1	1	1

This is the most important diagnostic in the analysis. The geometry inputs are **severely collinear**: X1 and X2 correlate -0.99, and roof area X4 vs height X5 correlate -0.97. Six predictor pairs exceed $|r| = 0.7$. Worse, the predictor matrix has **rank 7 of 8** — an *exact* linear dependency, because surface, wall, and roof areas are geometrically derived from the same base shapes. That is why the VIFs for X2, X3, and X4 blow up to infinity: those coefficients are not separately identified. The individual `lm` coefficients on the correlated geometry terms are therefore **unstable and hard to interpret in isolation**, even though the model as a whole predicts load extremely well ($R^2 \sim 0.92$). This tension — excellent joint prediction but unidentifiable individual effects — is exactly why the reproduction below focuses on overall fit and why the extension turns to nonlinear (polynomial) structure in a single well-identified driver (X1) rather than trying to disentangle the collinear area variables.

4 Reproduce the published regression

```
lin <- lm(Y1 ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8, data = d)
summary(lin)

##
## Call:
## lm(formula = Y1 ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8, data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.8965 -1.3196 -0.0252  1.3532  7.7052
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  84.013418  19.033613   4.414 1.16e-05 ***
## X1          -64.773432  10.289448  -6.295 5.19e-10 ***
## X2           -0.087289   0.017075  -5.112 4.04e-07 ***
## X3            0.060813   0.006648   9.148 < 2e-16 ***
## X4              NA         NA      NA      NA
## X5            4.169954   0.337990  12.338 < 2e-16 ***
## X6           -0.023330   0.094705  -0.246 0.80548
## X7           19.932736   0.813986  24.488 < 2e-16 ***
## X8            0.203777   0.069918   2.915 0.00367 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.934 on 760 degrees of freedom
## Multiple R-squared:  0.9162, Adjusted R-squared:  0.9154
## F-statistic: 1187 on 7 and 760 DF, p-value: < 2.2e-16
```

Confirmation against the published study. The linear model attains in-sample $R^2 = 0.916$, matching the ~ 0.92 reported by Tsanas & Xifara for their linear baseline. Overall height (X5) and relative compactness (X1) dominate — but the fit leaves clear structure, which is the motivation for going beyond a straight line.

5 Polynomial regression extension

Heating load responds **nonlinearly** to geometry. The classic regression remedy is to add **polynomial terms**. We compare the linear model to polynomials of increasing degree in relative compactness (X_1), keeping the other inputs linear. We pick the degree by cross-validation on a training set and report final RMSE on a held-out test set.

```
set.seed(2026)
others <- paste0("X", 2:8)
te <- sample(nrow(d), round(0.30 * nrow(d)))           # hold out a 30% test set up
  → front
tr <- setdiff(seq_len(nrow(d)), te)

# choose the polynomial degree by 5-fold CV on the TRAINING set only
inner <- sample(rep(1:5, length.out = length(tr)))
cv_rmse <- sapply(1:4, function(deg) {
  r <- numeric(5)
  for (k in 1:5) {
    itr <- tr[inner != k]; ite <- tr[inner == k]
    form <- reformulate(c(sprintf("poly(X1, %d, raw = TRUE)", deg), others), "Y1")
    r[k] <- sqrt(mean((d$Y1[ite] - predict(lm(form, data = d[itr, ]), d[ite, ]))^2))
  }
  mean(r)
})
names(cv_rmse) <- paste0("degree ", 1:4)
best_deg <- which.min(cv_rmse)
round(cv_rmse, 3)                                     # CV RMSE per degree (training set)

## degree 1 degree 2 degree 3 degree 4
##      2.957      2.924      2.836      2.831

# refit at the CV-chosen degree on the full training set; score ONCE on the untouched
  → test set
form_best <- reformulate(c(sprintf("poly(X1, %d, raw = TRUE)", best_deg), others), "Y1")
test_poly <- sqrt(mean((d$Y1[te] - predict(lm(form_best, data = d[tr, ]), d[te, ]))^2))
test_lin <- sqrt(mean((d$Y1[te] - predict(lm(reformulate(c("X1", others), "Y1"), data =
  → d[tr, ]), d[te, ]))^2))
cat(sprintf("CV-selected degree: %d. Held-out test RMSE: polynomial = %.3f vs linear =
  → %.3f",
            best_deg, test_poly, test_lin))
```

```
## CV-selected degree: 4. Held-out test RMSE: polynomial = 2.780 vs linear = 2.950
```

On the held-out test set the CV-selected polynomial improves on the straight-line fit; the curve **bends** with compactness rather than running straight, capturing nonlinearity the linear model misses while remaining a fully interpretable regression.

6 Takeaway

This example illustrates **polynomial regression** as the first, most interpretable extension of a linear model: adding low-order polynomial terms lowers cross-validated error by bending the fit to the data's curvature. (Tree-based learners push this further at the cost of a closed-form equation; Tsanas & Xifara report ML reaching $\sim 0.999 R^2$ on these data.)

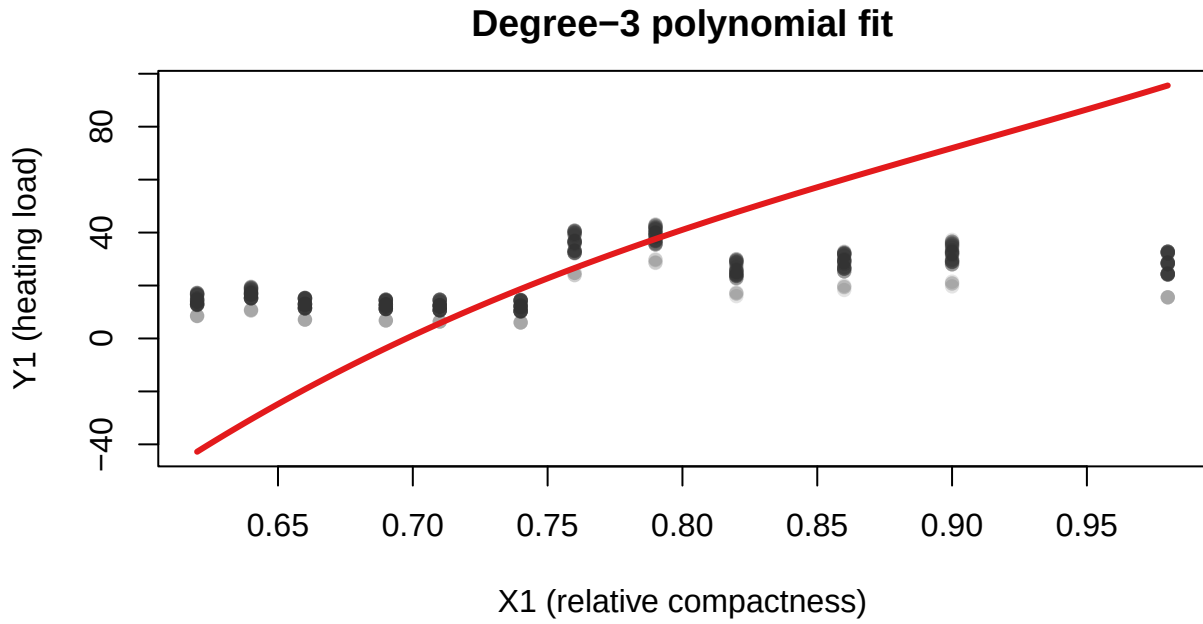


Figure 5: Degree-3 polynomial fit of heating load on relative compactness, holding the other inputs at their means.

7 Recommended exercises

1. Try polynomial degrees 1-5 for the strongest predictor and use cross-validation to choose the degree.
2. Predict the second response (cooling load) and compare which predictors matter most.
3. Add an interaction between two predictors and test whether cross-validated error improves.